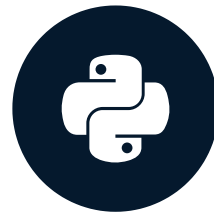


Count plots and bar plots

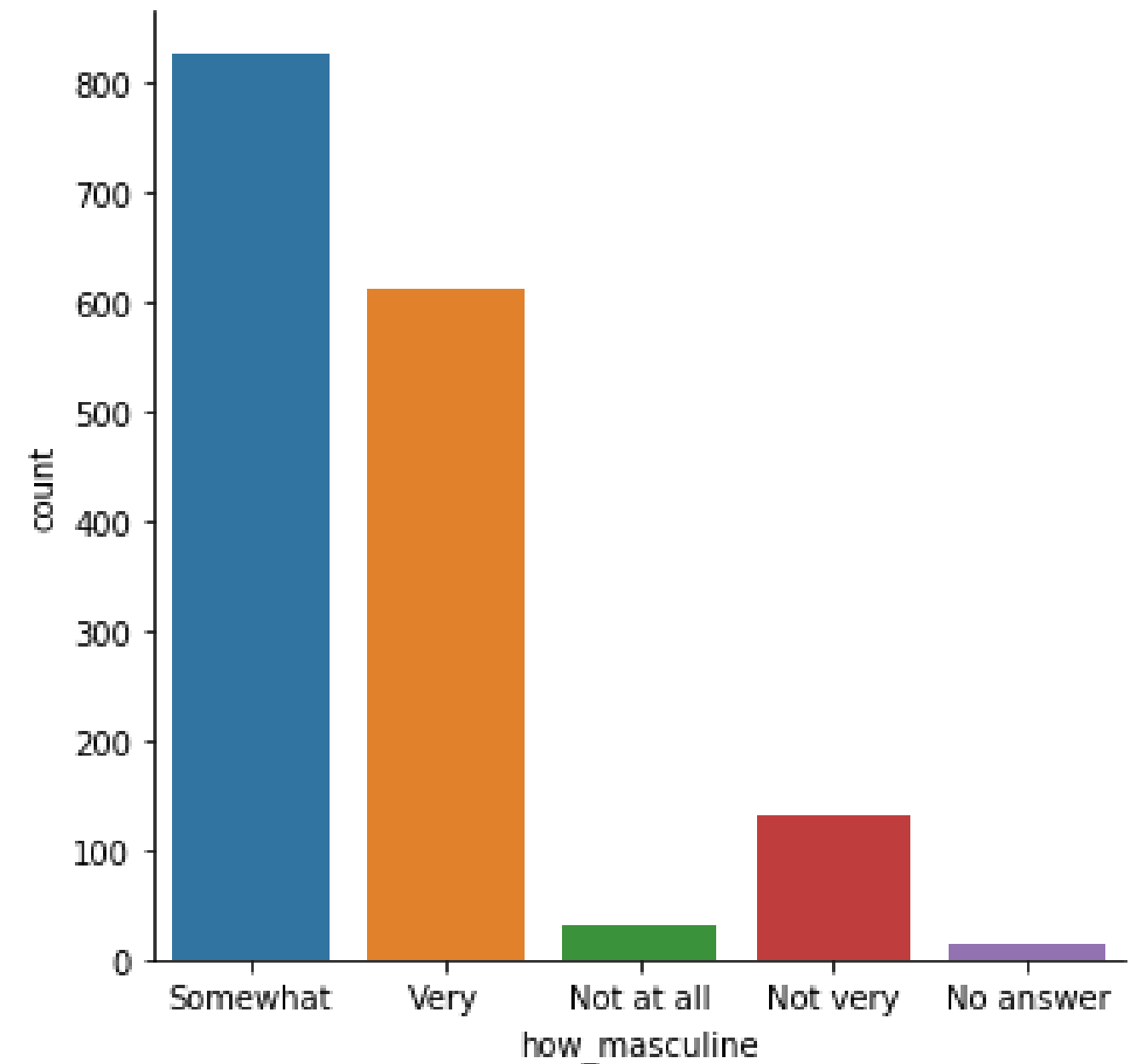
INTRODUCTION TO DATA VISUALIZATION WITH SEABORN



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Categorical plots

- Examples: count plots, bar plots
- Involve a categorical variable
- Comparisons between groups



catplot()

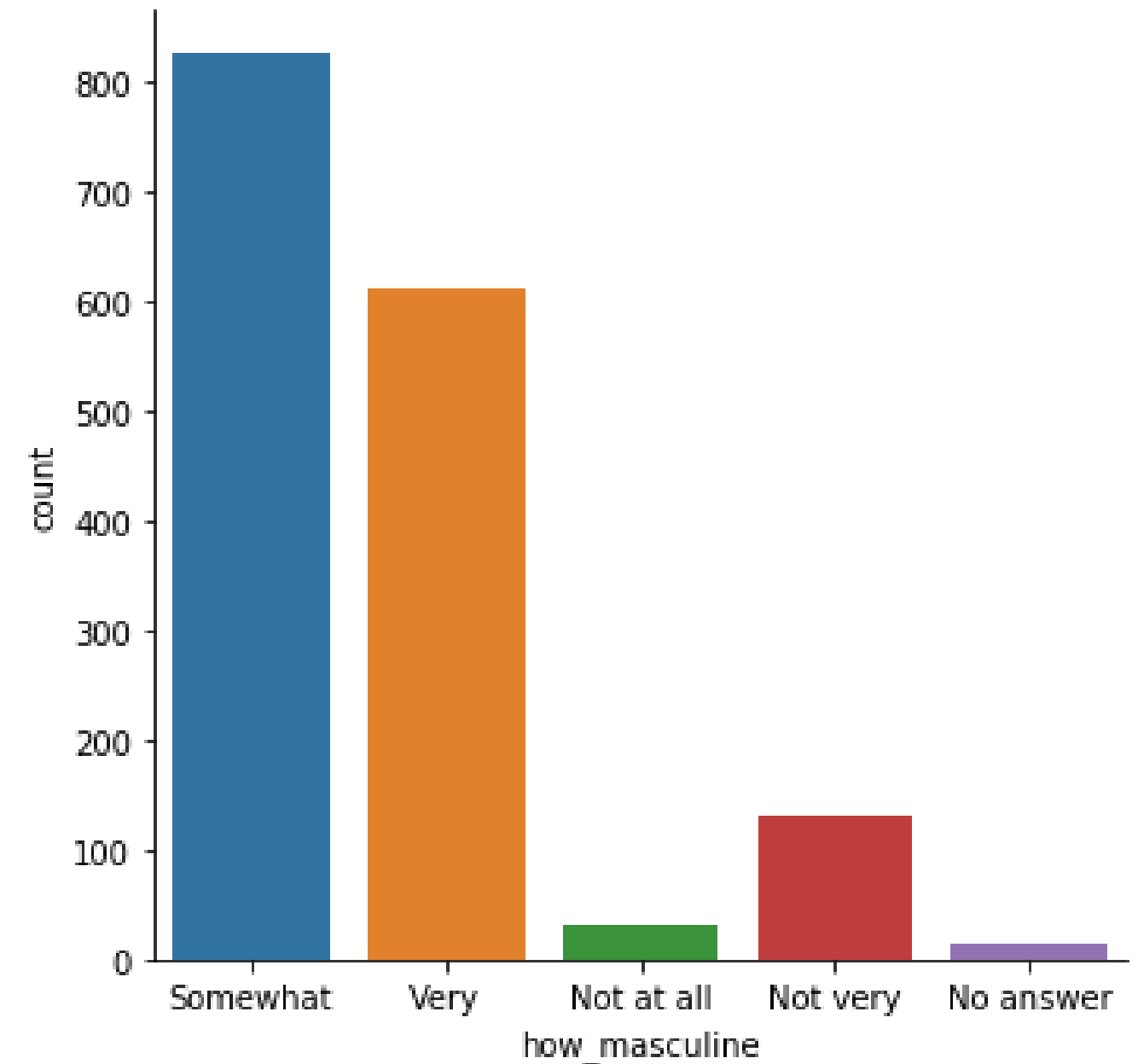
- Used to create categorical plots
- Same advantages of `relplot()`
- Easily create subplots with `col=` and `row=`

countplot() vs. catplot()

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.countplot(x="how_masculine",
              data=masculinity_data)

plt.show()
```

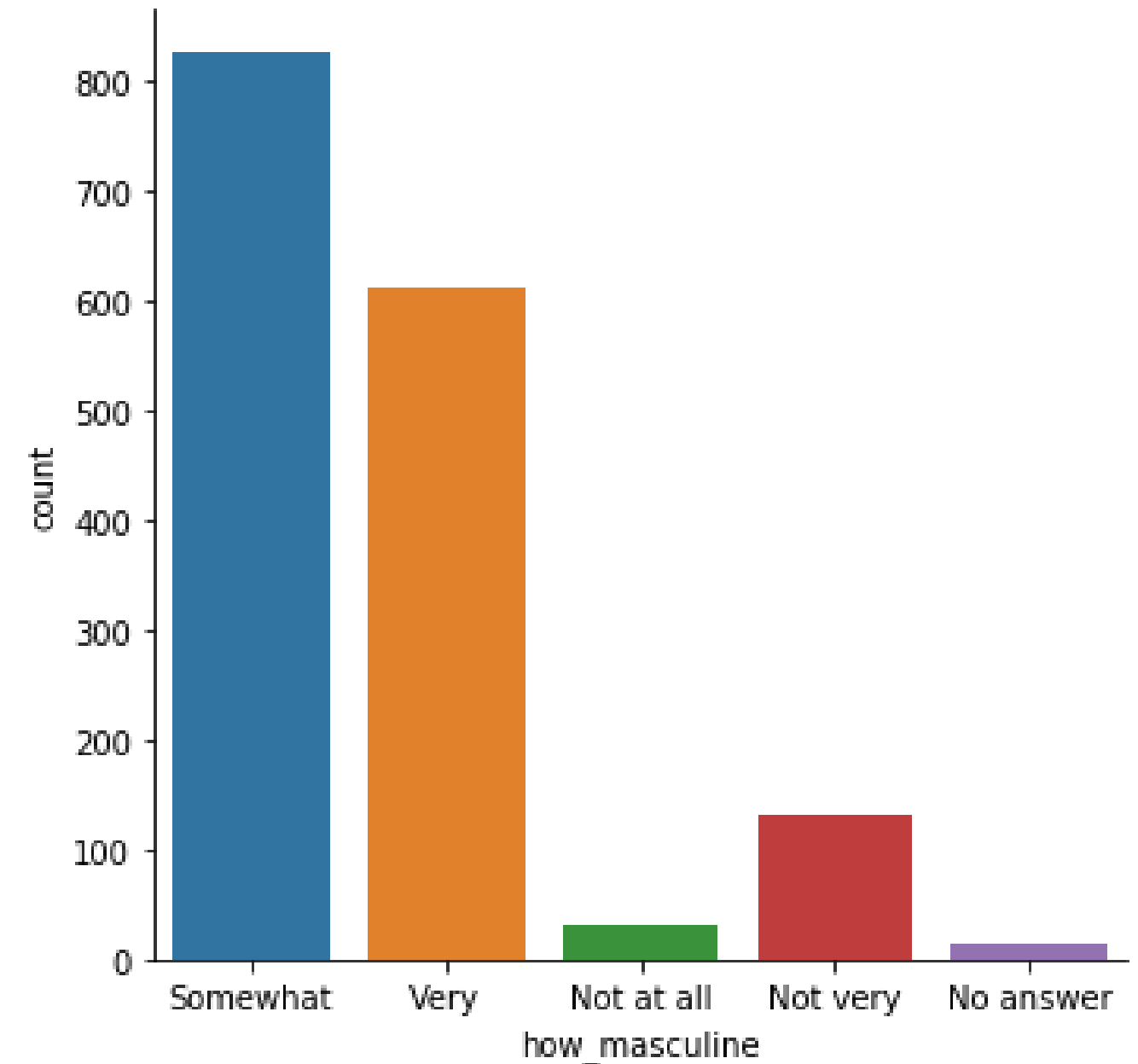


countplot() vs. catplot()

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="how_masculine",
            data=masculinity_data,
            kind="count")

plt.show()
```

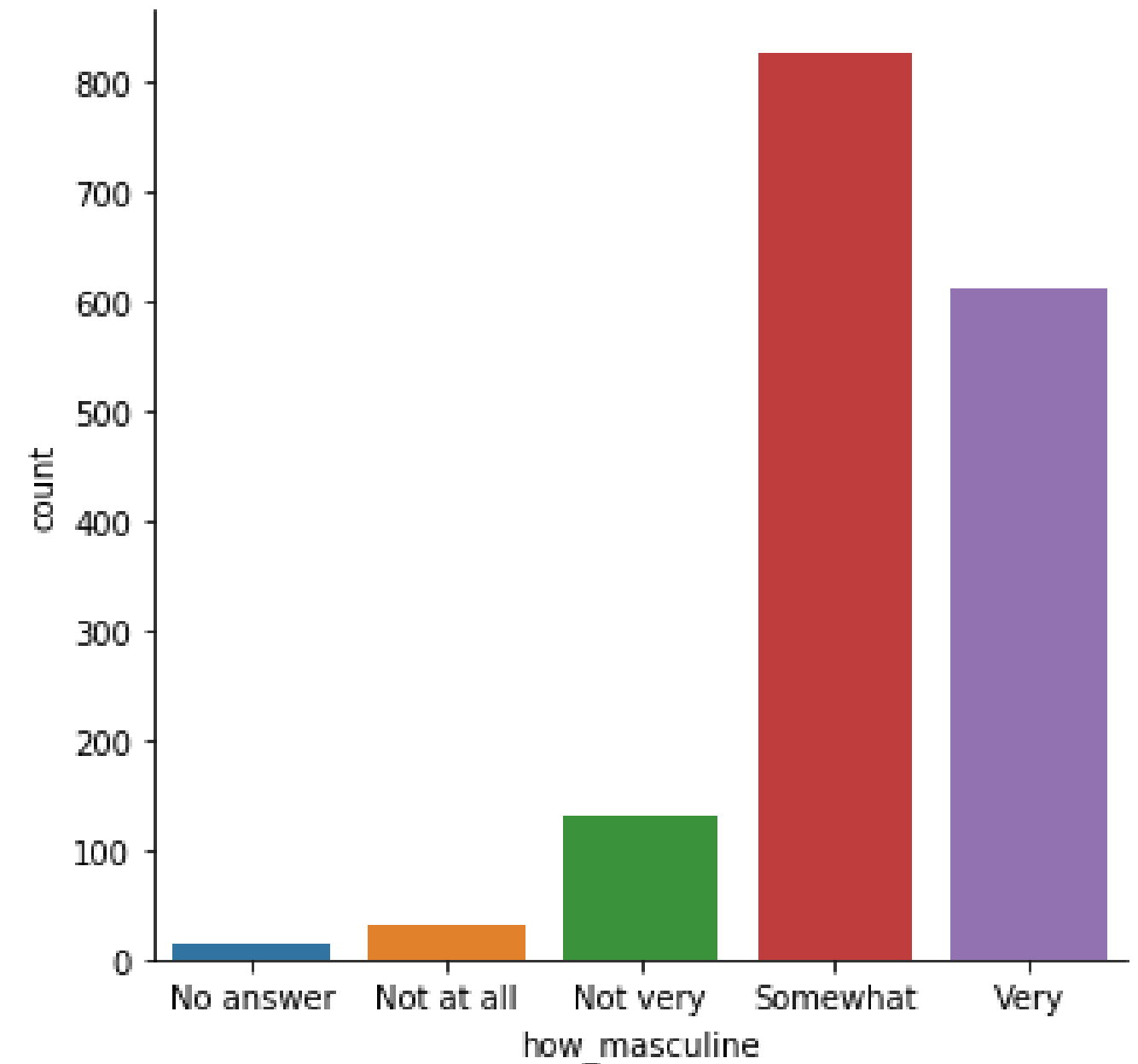


Changing the order

```
import matplotlib.pyplot as plt
import seaborn as sns
category_order = ["No answer",
                  "Not at all",
                  "Not very",
                  "Somewhat",
                  "Very"]

sns.catplot(x="how_masculine",
            data=masculinity_data,
            kind="count",
            order=category_order)

plt.show()
```



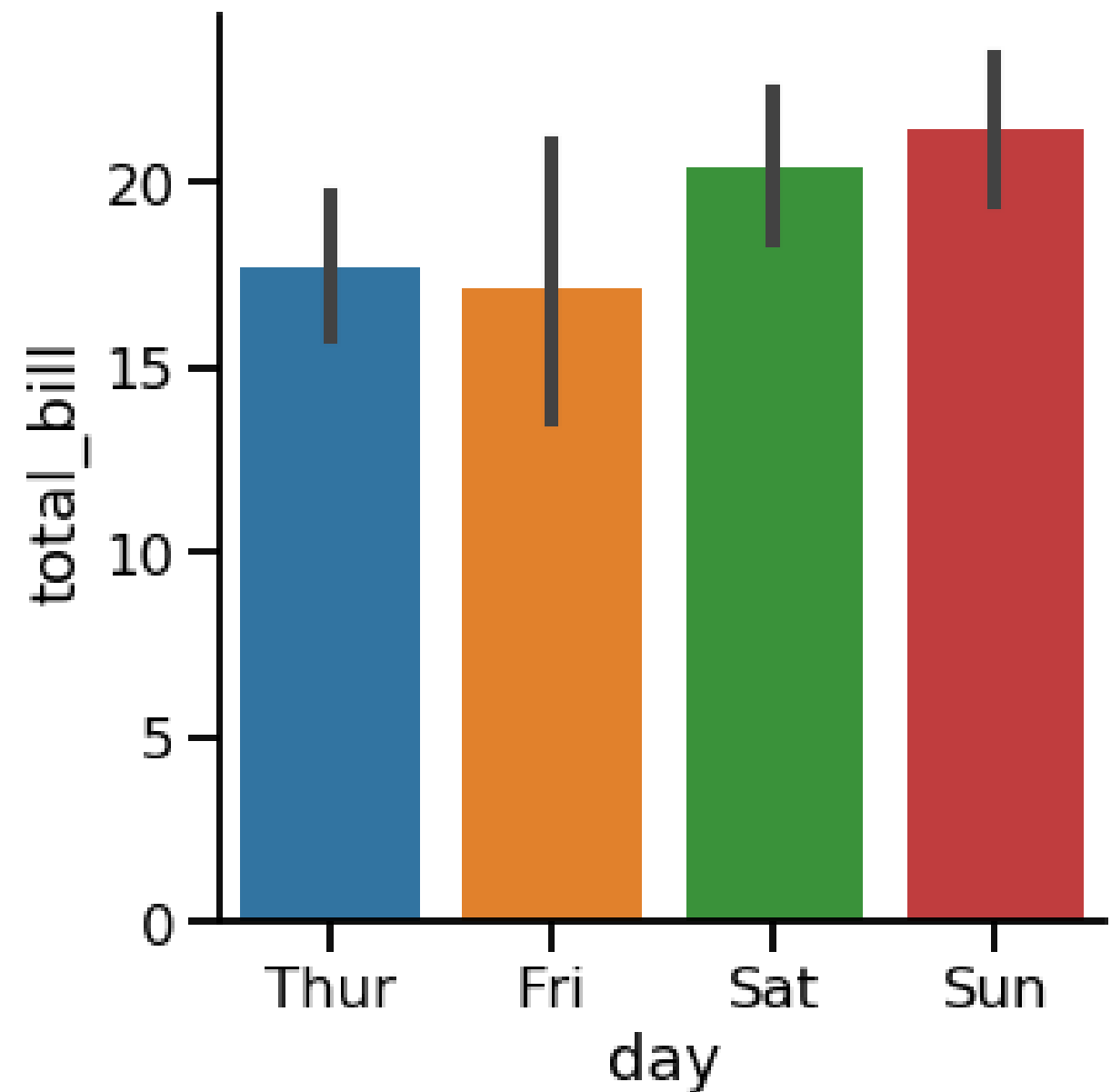
Bar plots

Displays mean of quantitative variable per category

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="day",
            y="total_bill",
            data=tips,
            kind="bar")

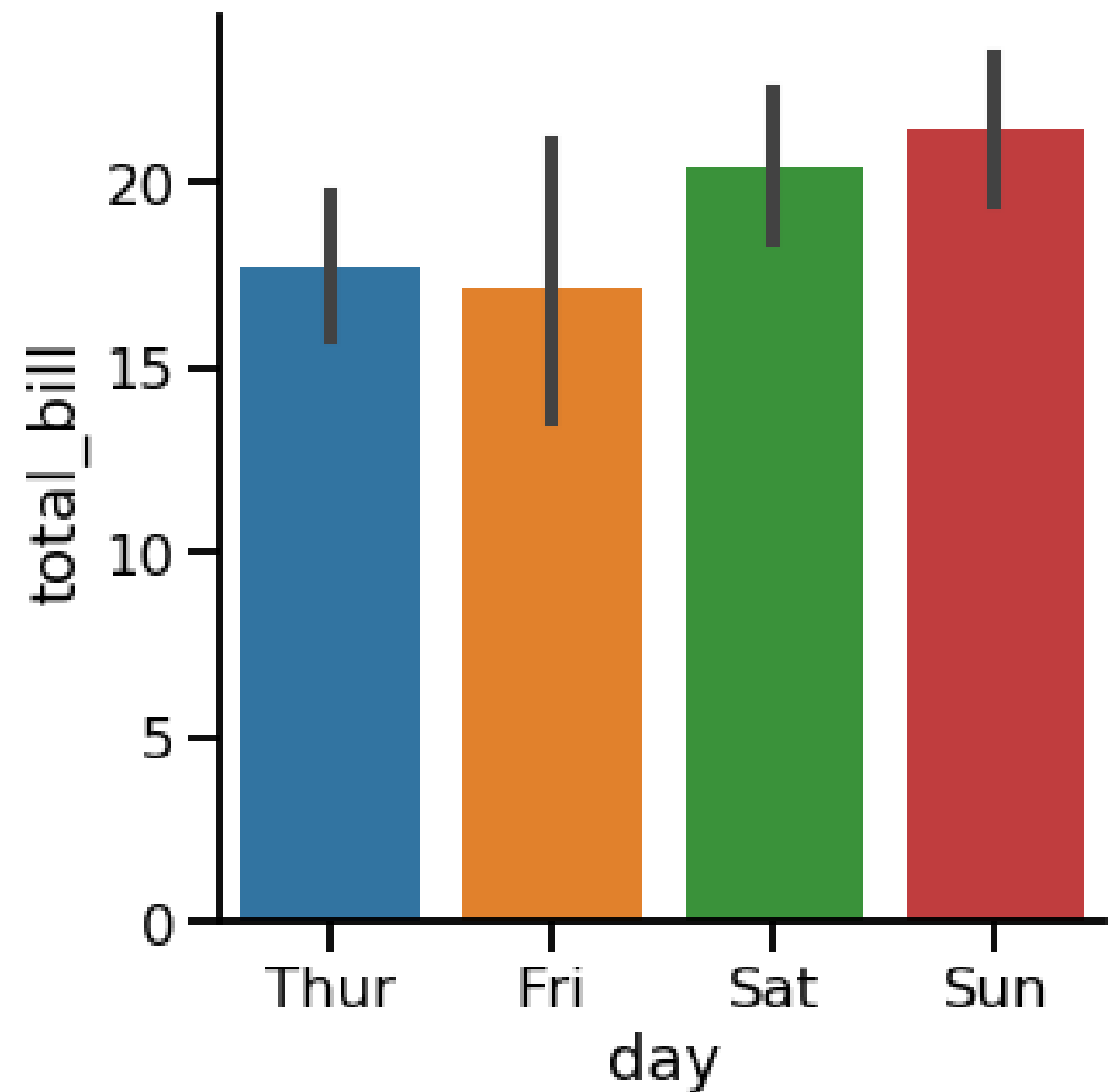
plt.show()
```



¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Confidence intervals

- Lines show 95% confidence intervals for the mean
- Shows uncertainty about our estimate
- Assumes our data is a random sample



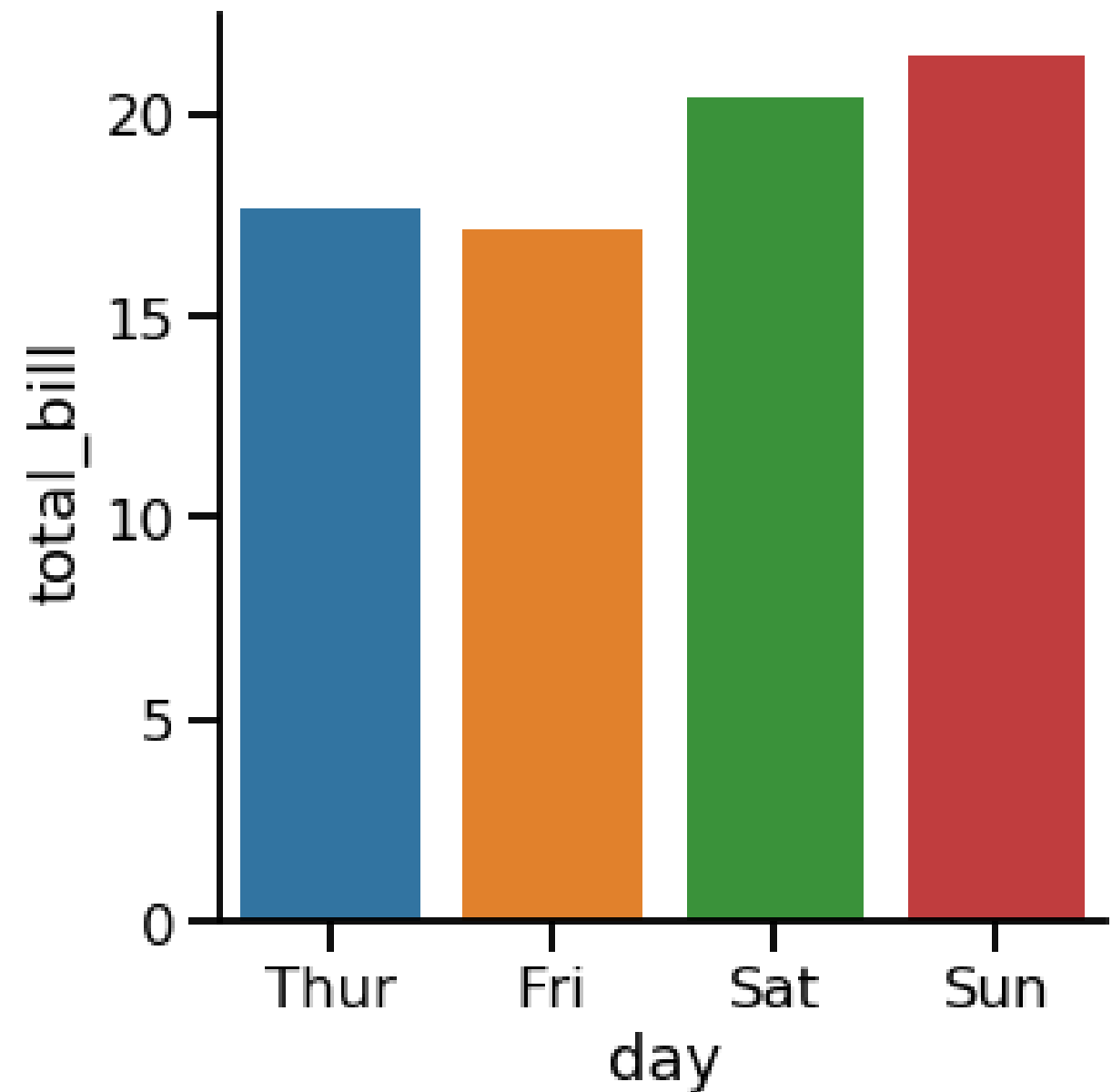
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Turning off confidence intervals

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="day",
            y="total_bill",
            data=tips,
            kind="bar",
            errorbar=None)

plt.show()
```



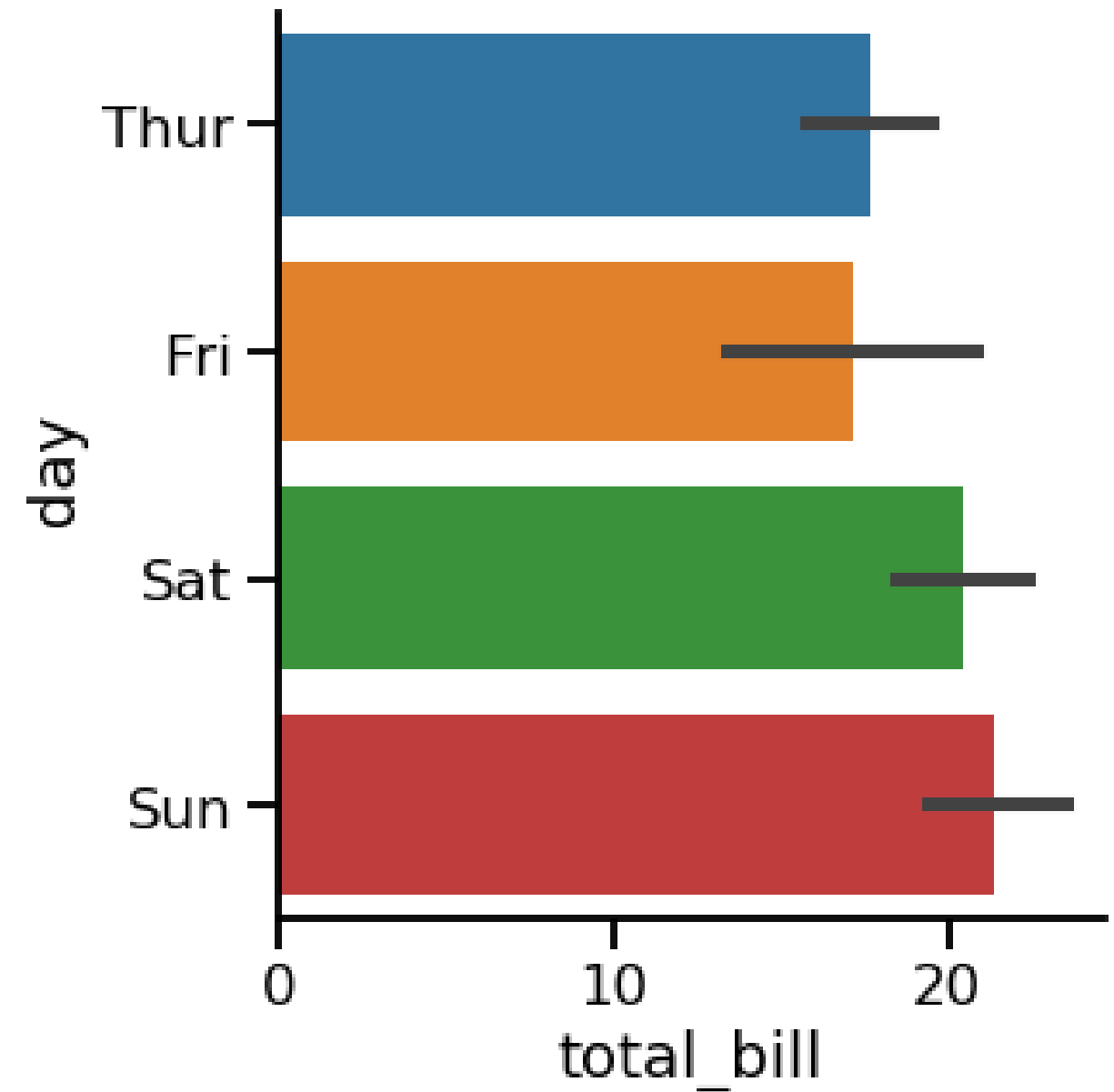
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Changing the orientation

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="total_bill",
            y="day",
            data=tips,
            kind="bar")

plt.show()
```



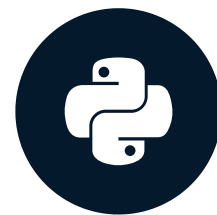
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Let's practice!

INTRODUCTION TO DATA VISUALIZATION WITH SEABORN

Creating a box plot

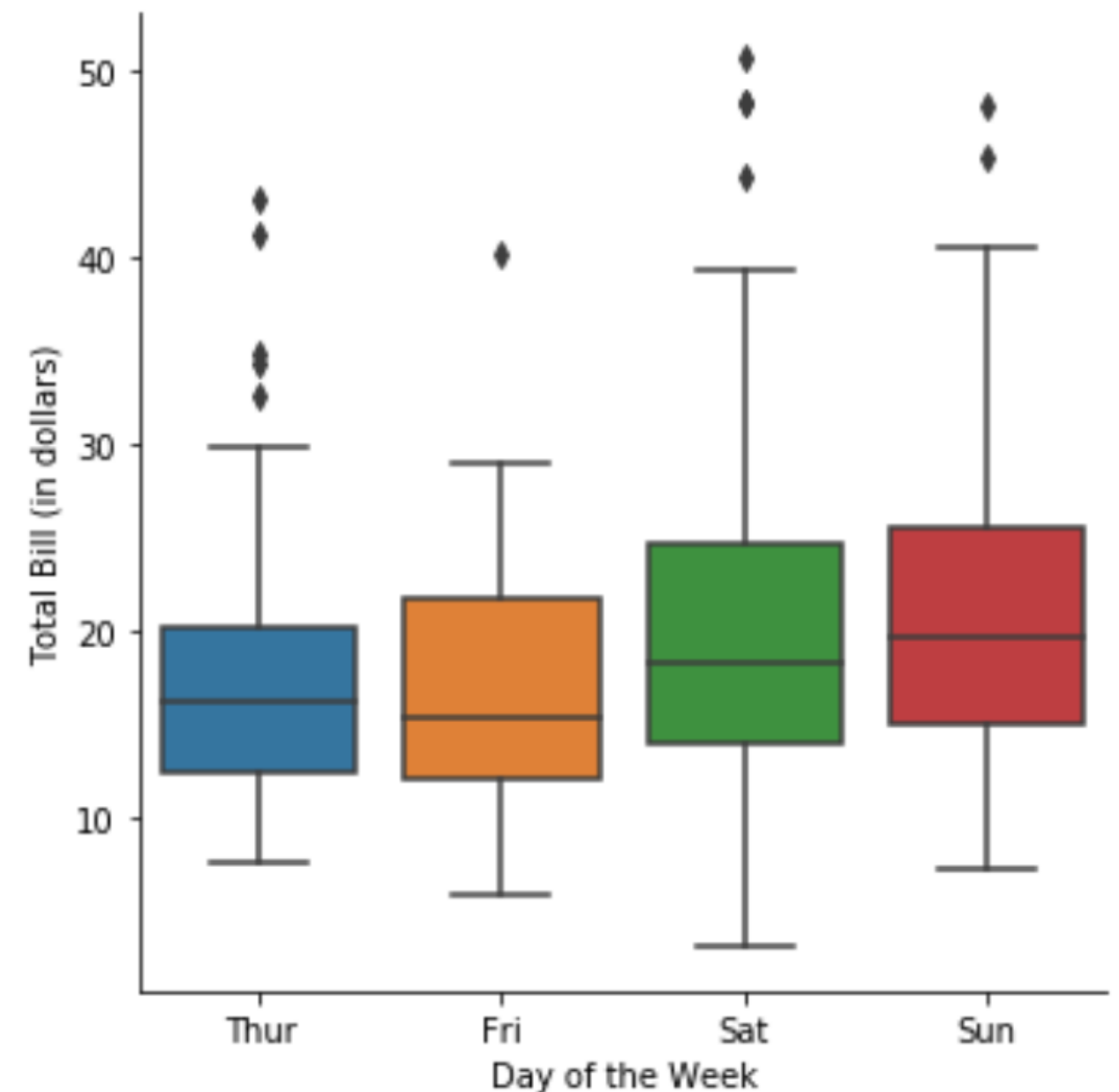
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What is a box plot?

- Shows the distribution of quantitative data
- See median, spread, skewness, and outliers
- Facilitates comparisons between groups



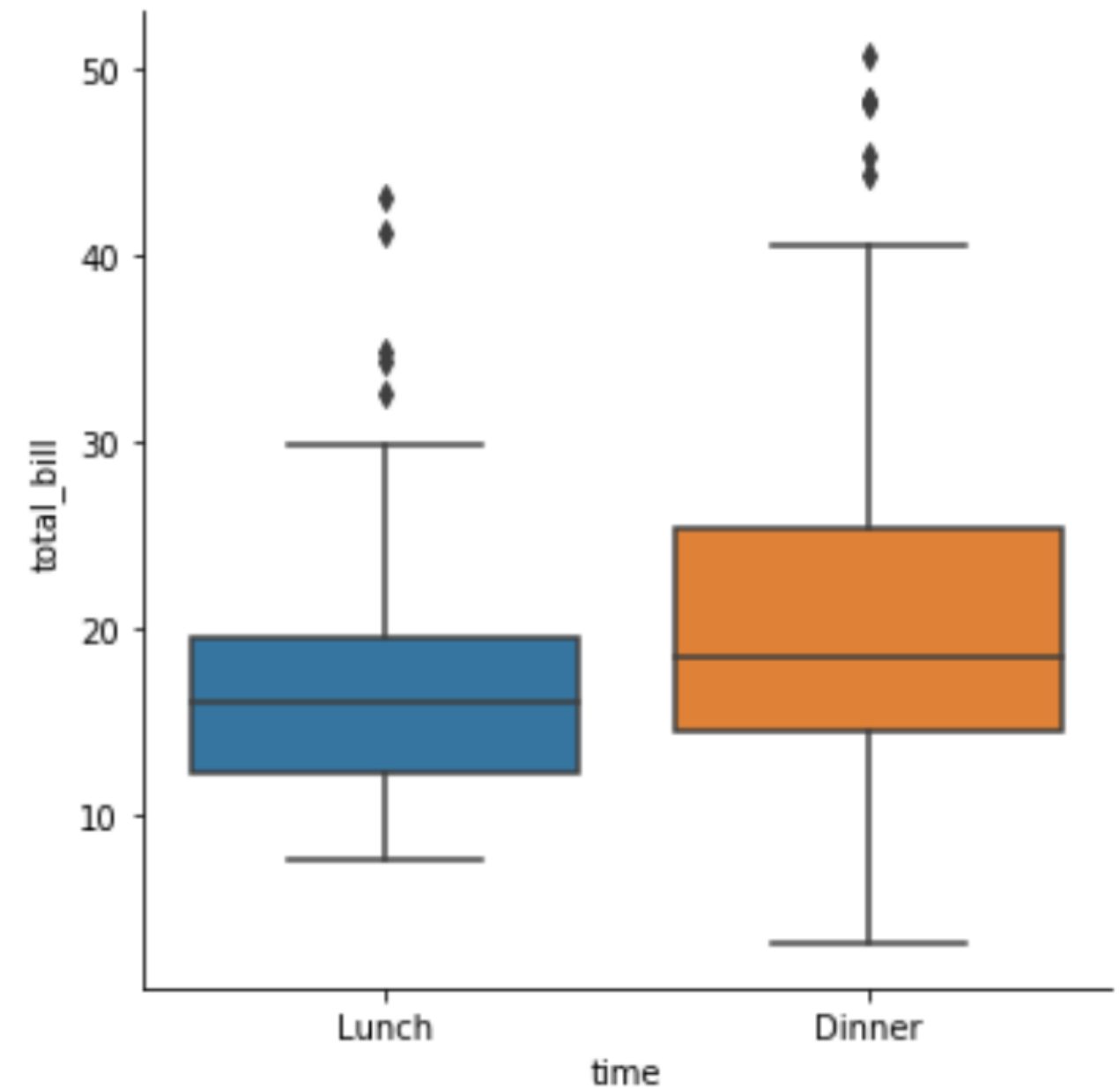
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

How to create a box plot

```
import matplotlib.pyplot as plt
import seaborn as sns

g = sns.catplot(x="time",
                y="total_bill",
                data=tips,
                kind="box")

plt.show()
```



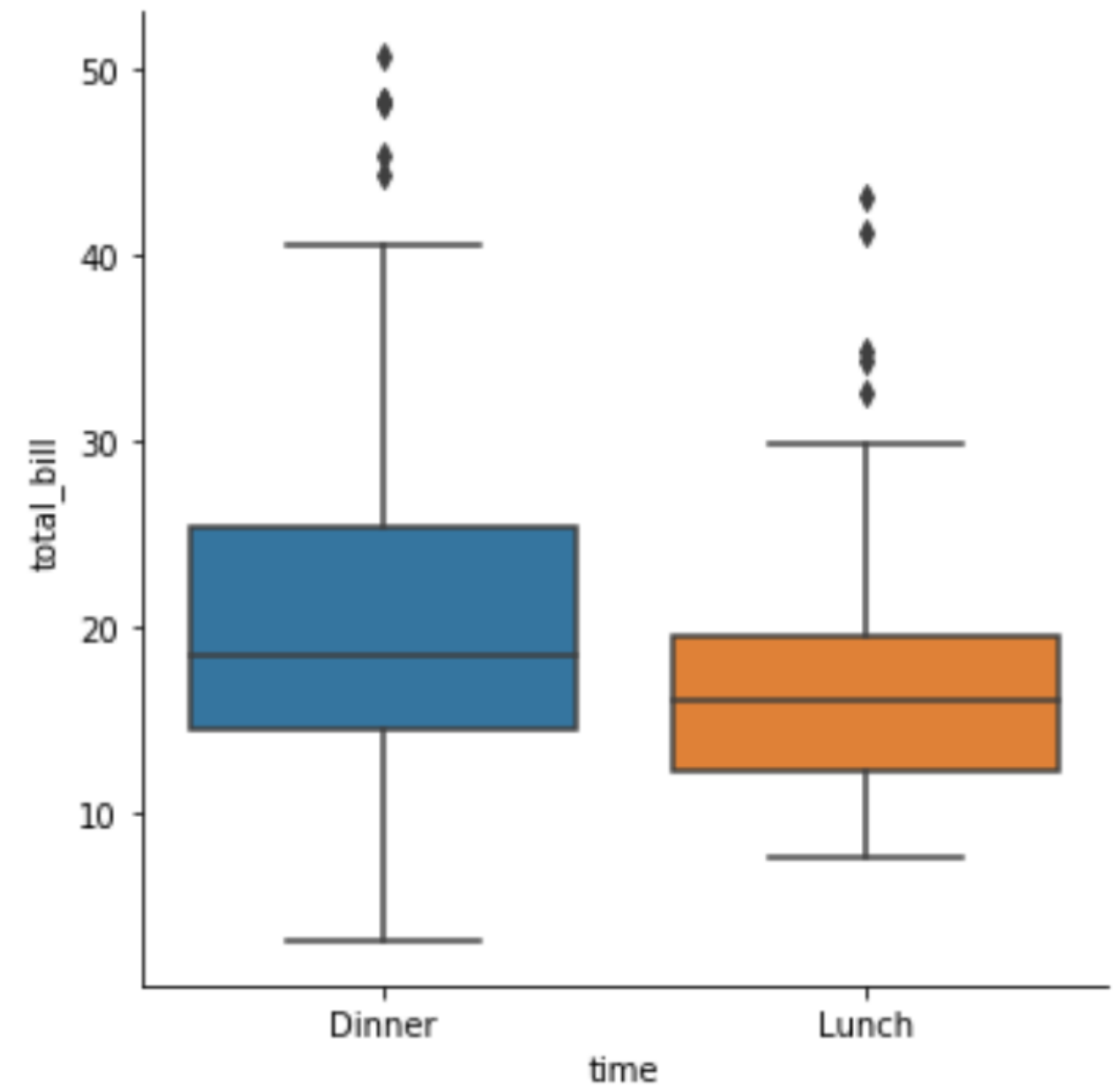
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Change the order of categories

```
import matplotlib.pyplot as plt
import seaborn as sns

g = sns.catplot(x="time",
                y="total_bill",
                data=tips,
                kind="box",
                order=["Dinner",
                     "Lunch"])

plt.show()
```



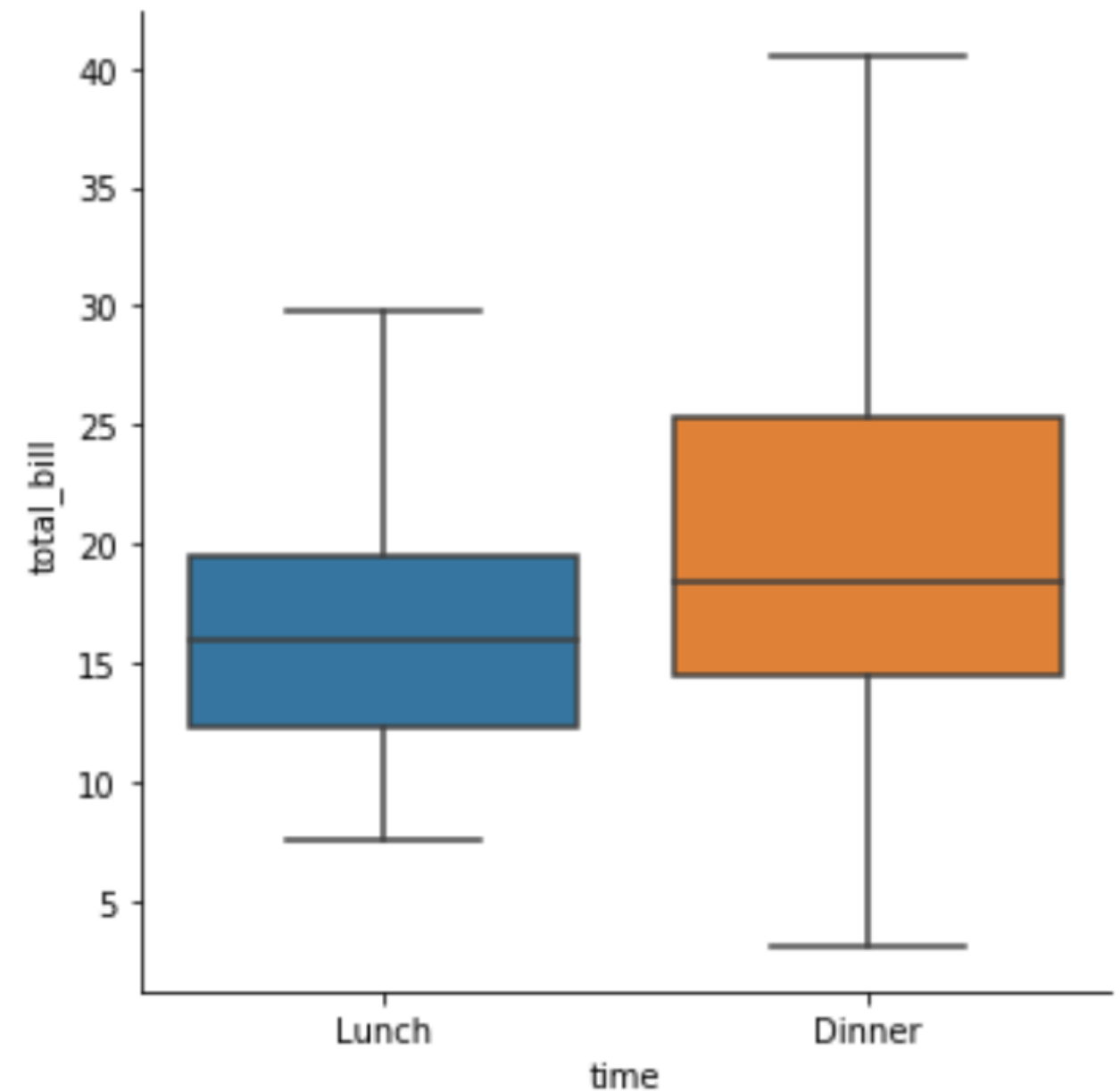
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Omitting the outliers

```
import matplotlib.pyplot as plt
import seaborn as sns

g = sns.catplot(x="time",
                y="total_bill",
                data=tips,
                kind="box",
                showfliers=False)

plt.show()
```



¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Changing the whiskers using `whis`

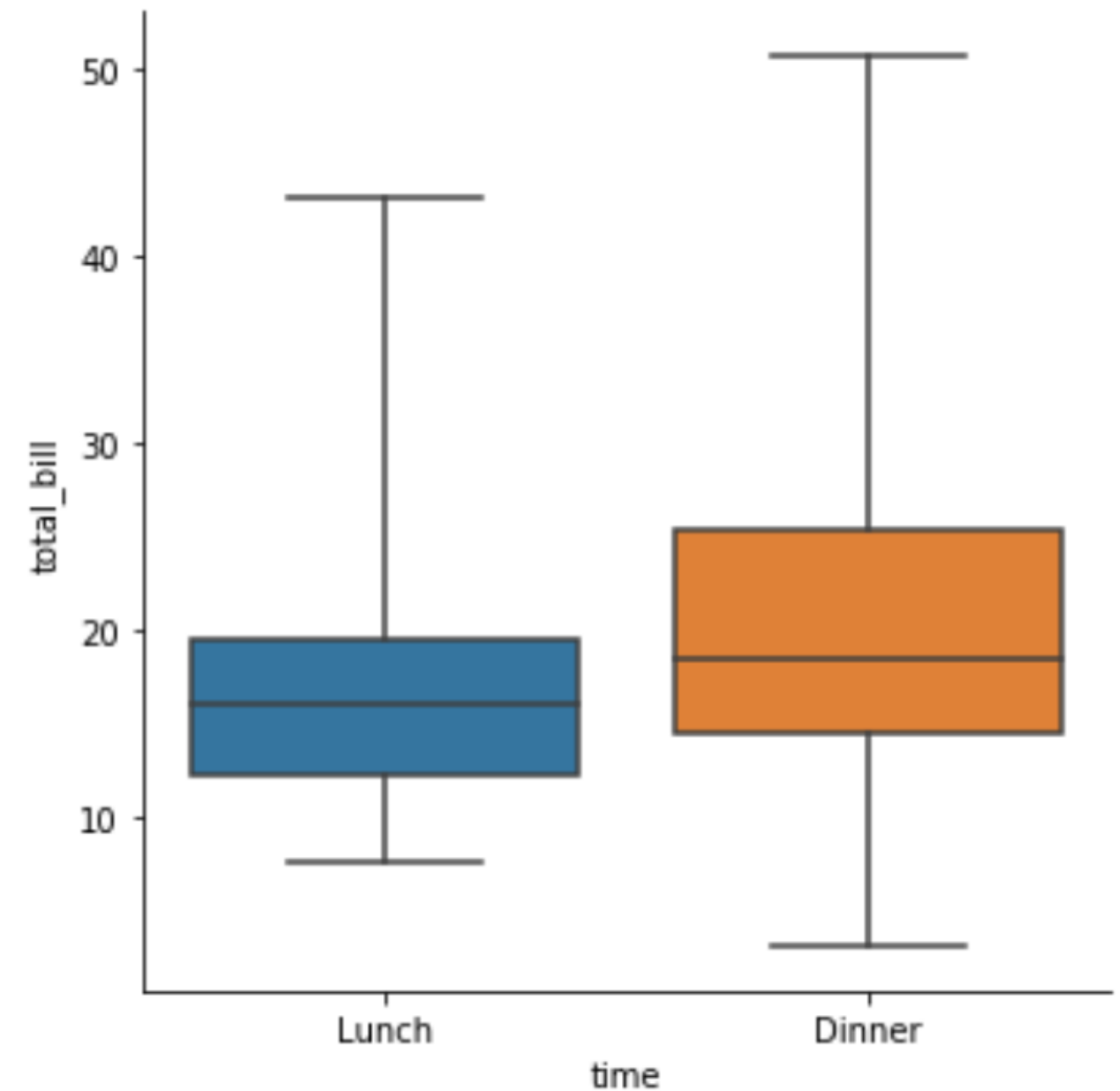
- By default, the whiskers extend to $1.5 \times$ the interquartile range
- Make them extend to $2.0 \times$ IQR: `whis=2.0`
- Show the 5th and 95th percentiles: `whis=[5, 95]`
- Show min and max values: `whis=[0, 100]`

Changing the whiskers using `whis`

```
import matplotlib.pyplot as plt
import seaborn as sns

g = sns.catplot(x="time",
                y="total_bill",
                data=tips,
                kind="box",
                whis=[0, 100])

plt.show()
```



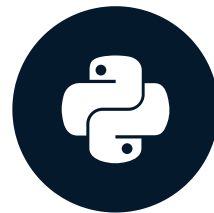
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Let's practice!

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Point plots

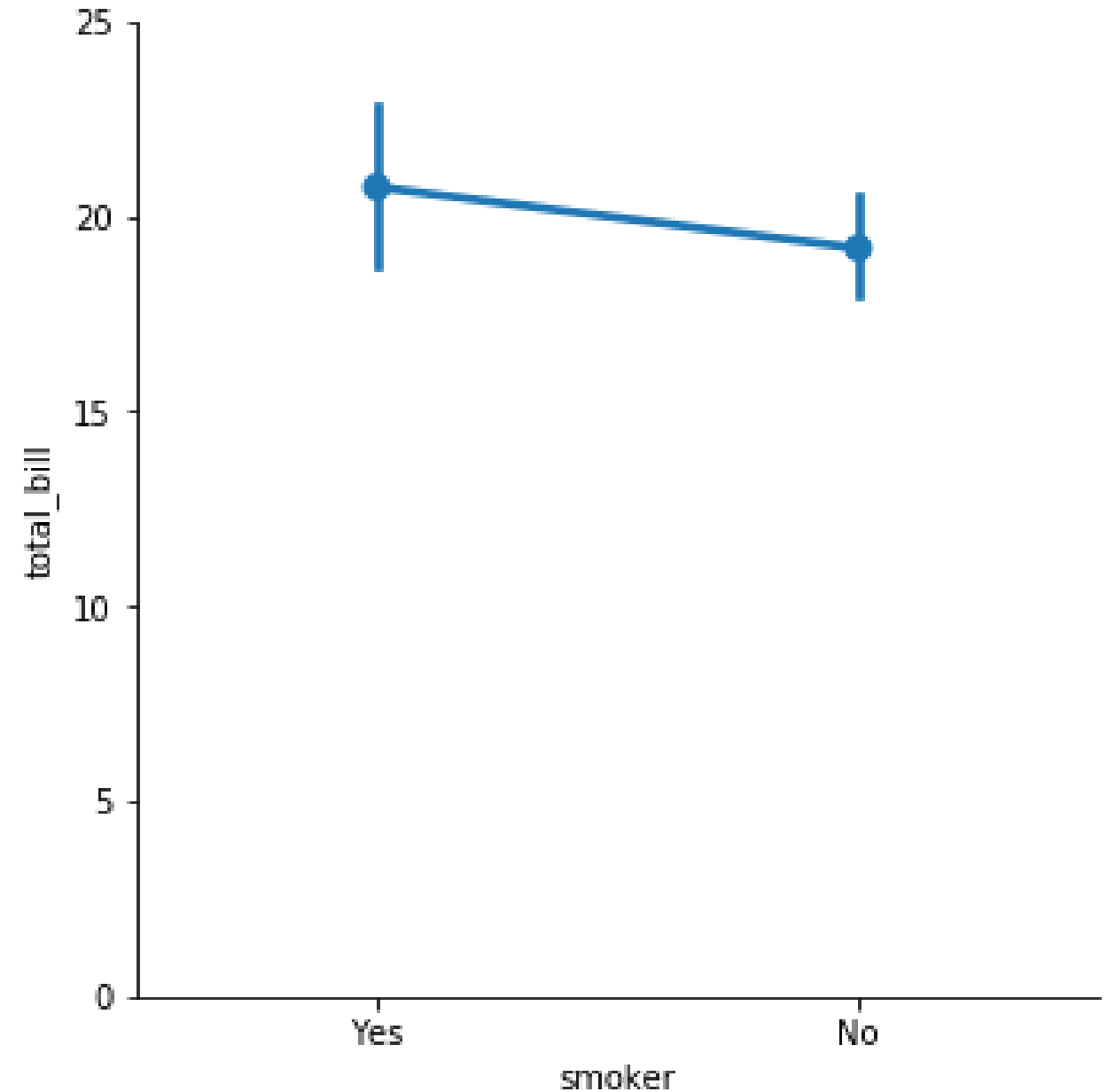
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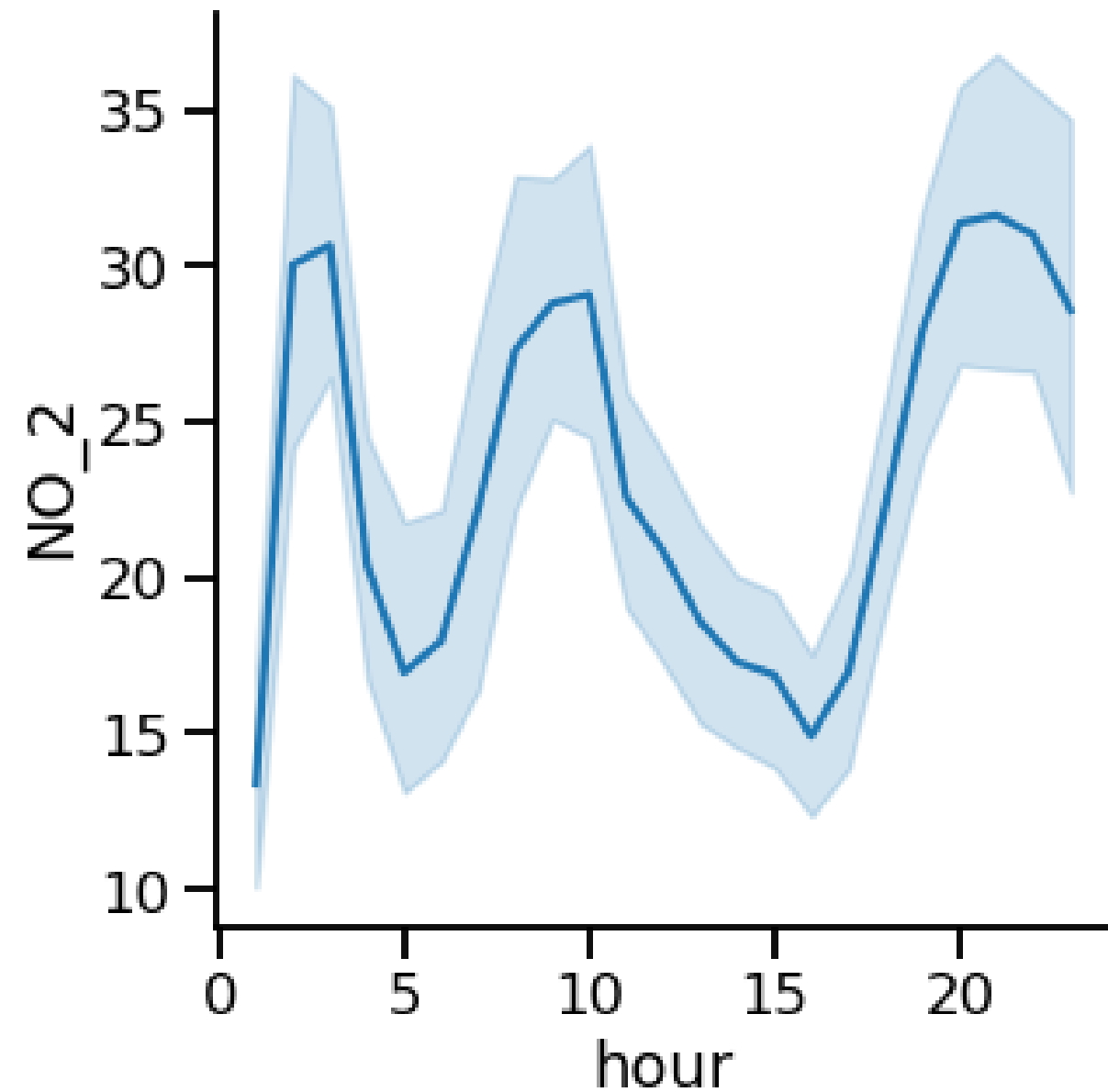
What are point plots?

- Points show mean of quantitative variable
- Vertical lines show 95% confidence intervals

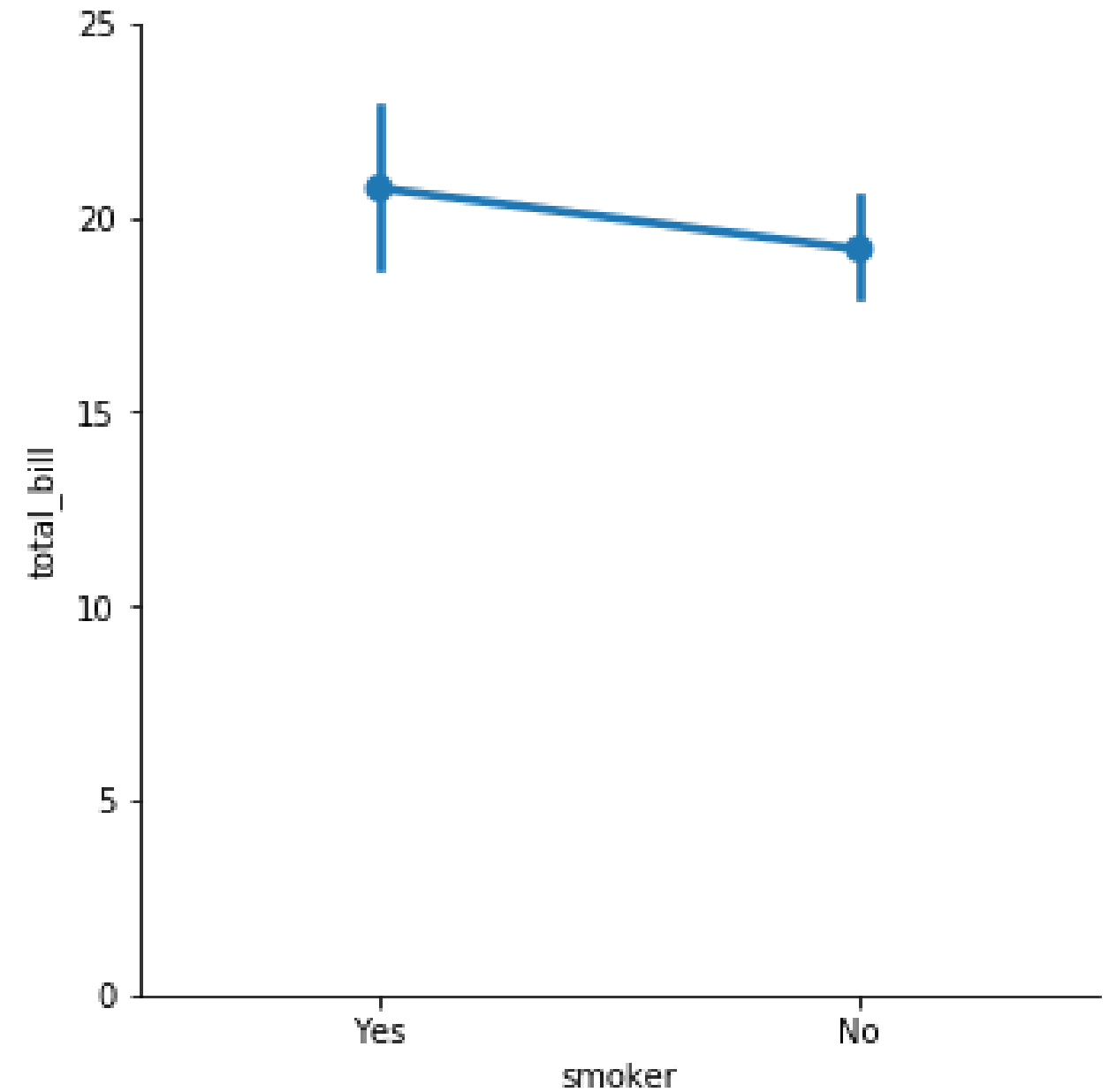


¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Line plot: average level of nitrogen dioxide over time



Point plot: average restaurant bill, smokers vs. non-smokers



¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Point plots vs. line plots

Both show:

- Mean of quantitative variable
- 95% confidence intervals for the mean

Differences:

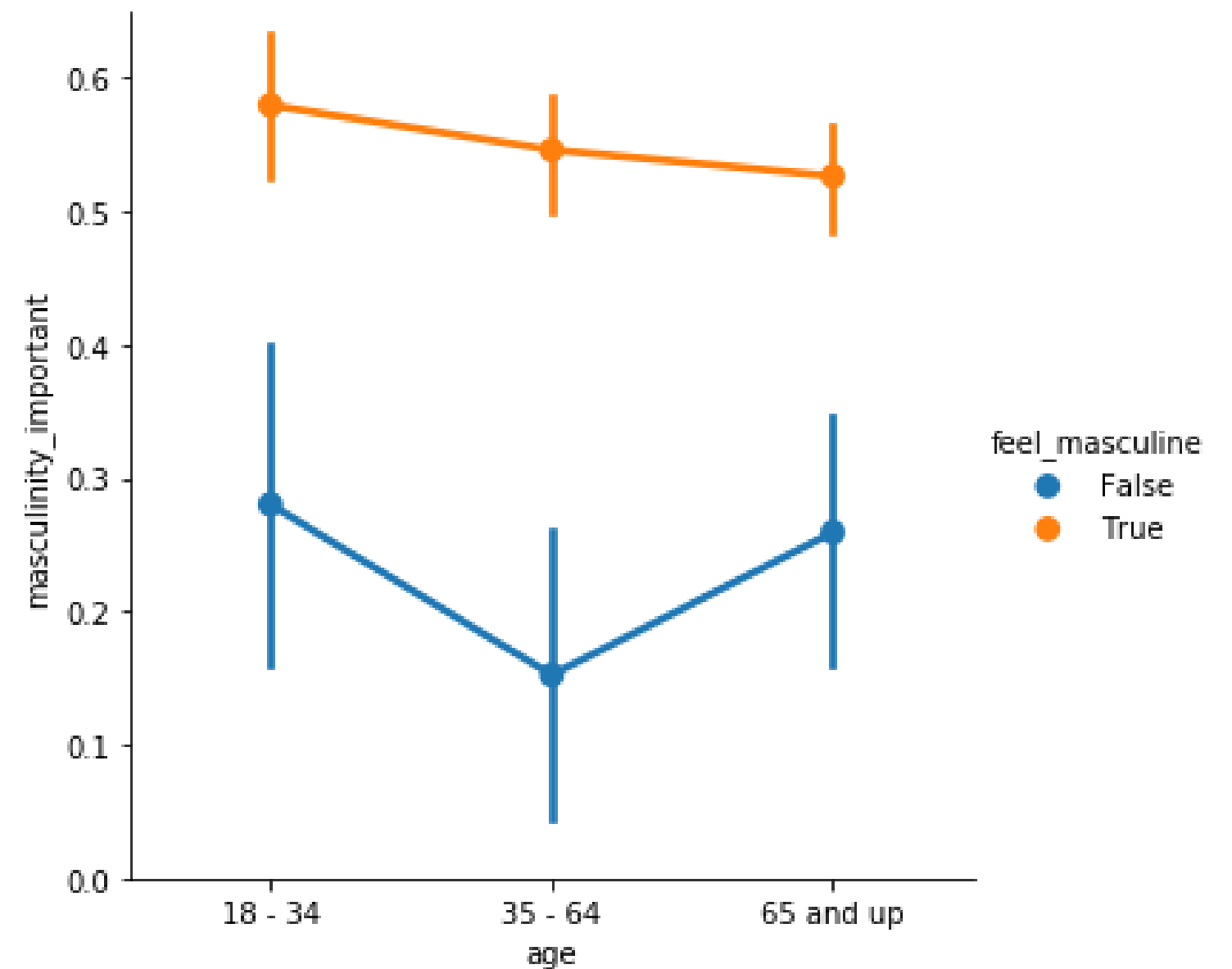
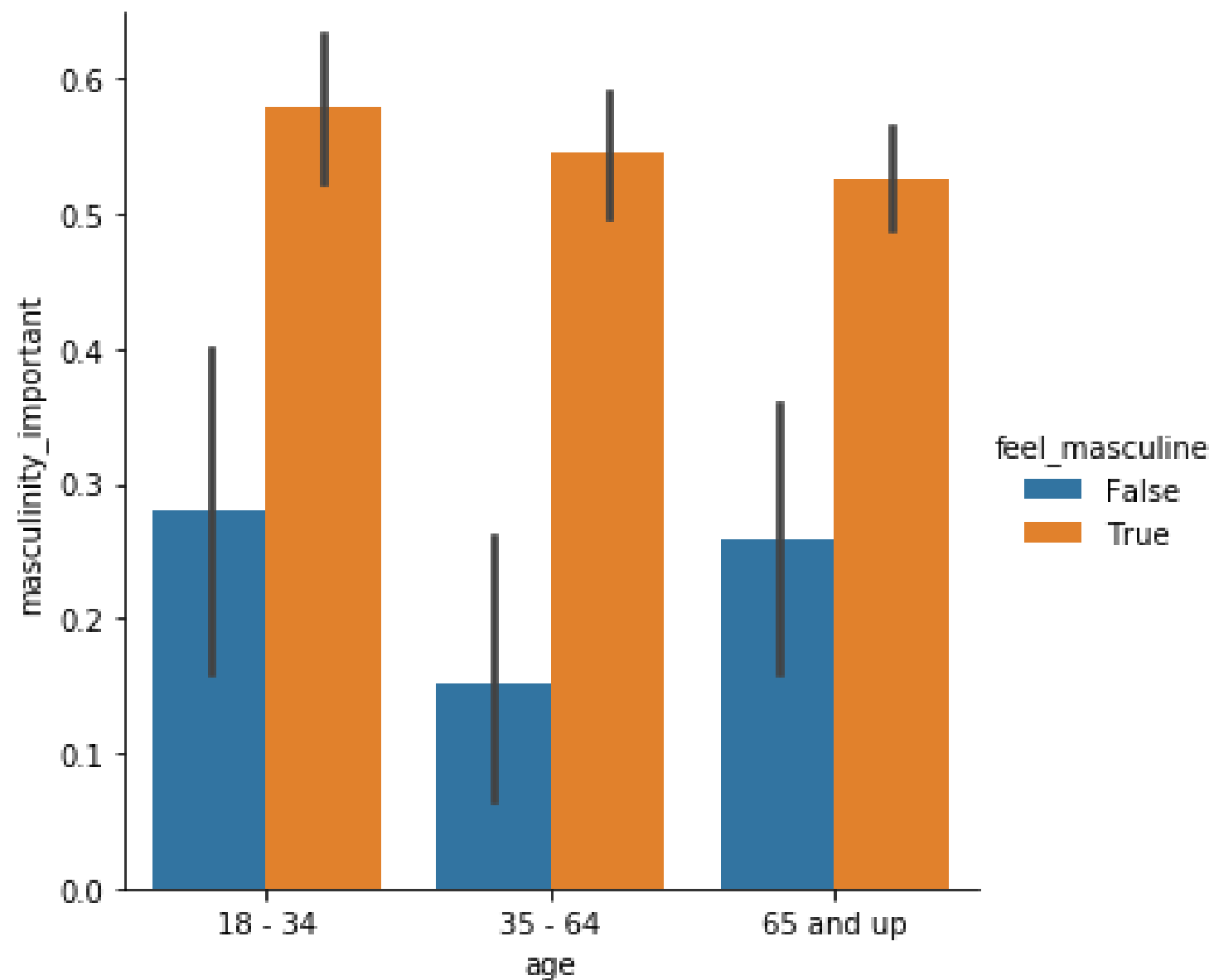
- Line plot has **quantitative** variable (usually time) on x-axis
- Point plot has **categorical** variable on x-axis

Point plots vs. bar plots

Both show:

- Mean of quantitative variable
- 95% confidence intervals for the mean

Point plots vs. bar plots

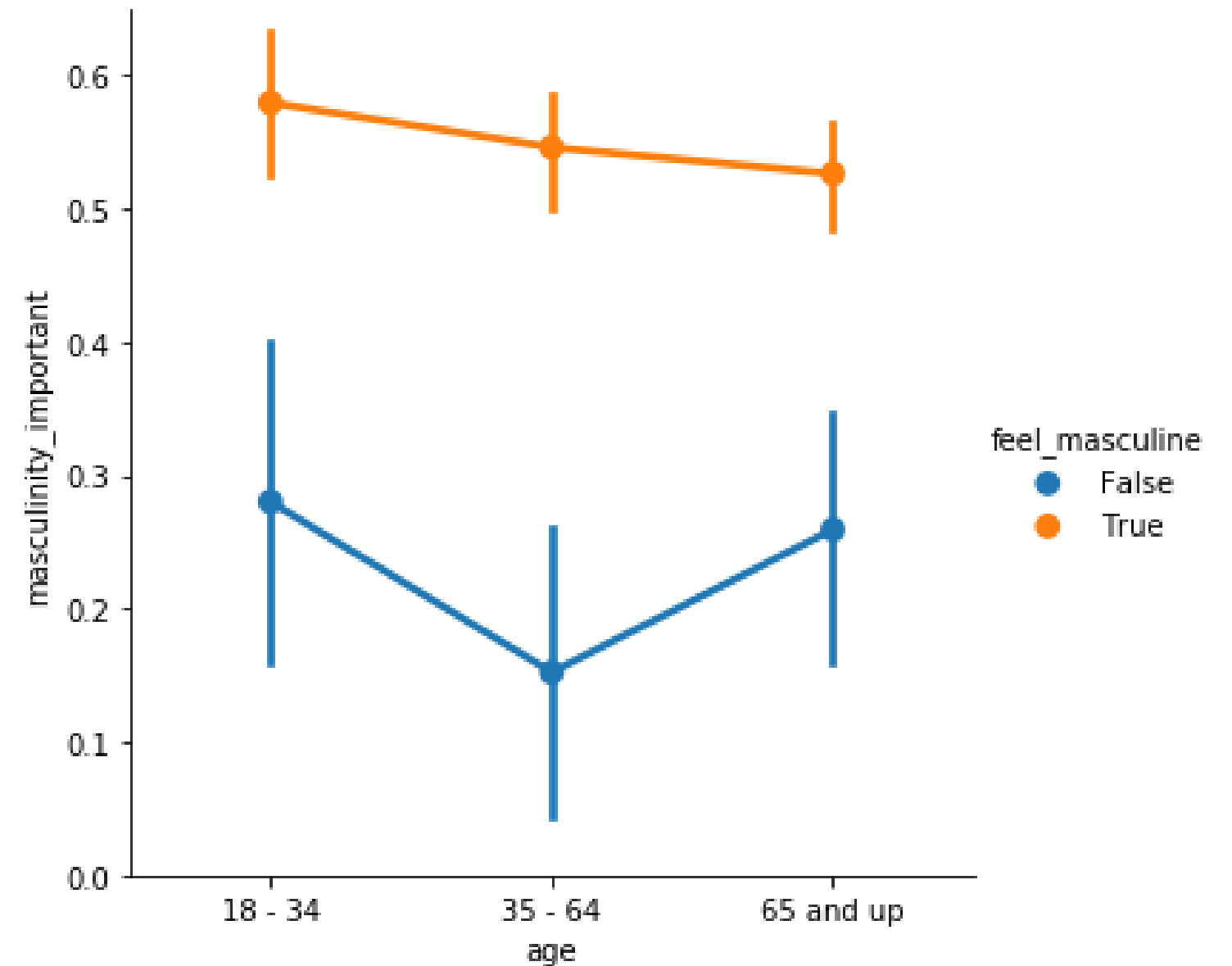


Creating a point plot

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="age",
            y="masculinity_important",
            data=masculinity_data,
            hue="feel_masculine",
            kind="point")

plt.show()
```

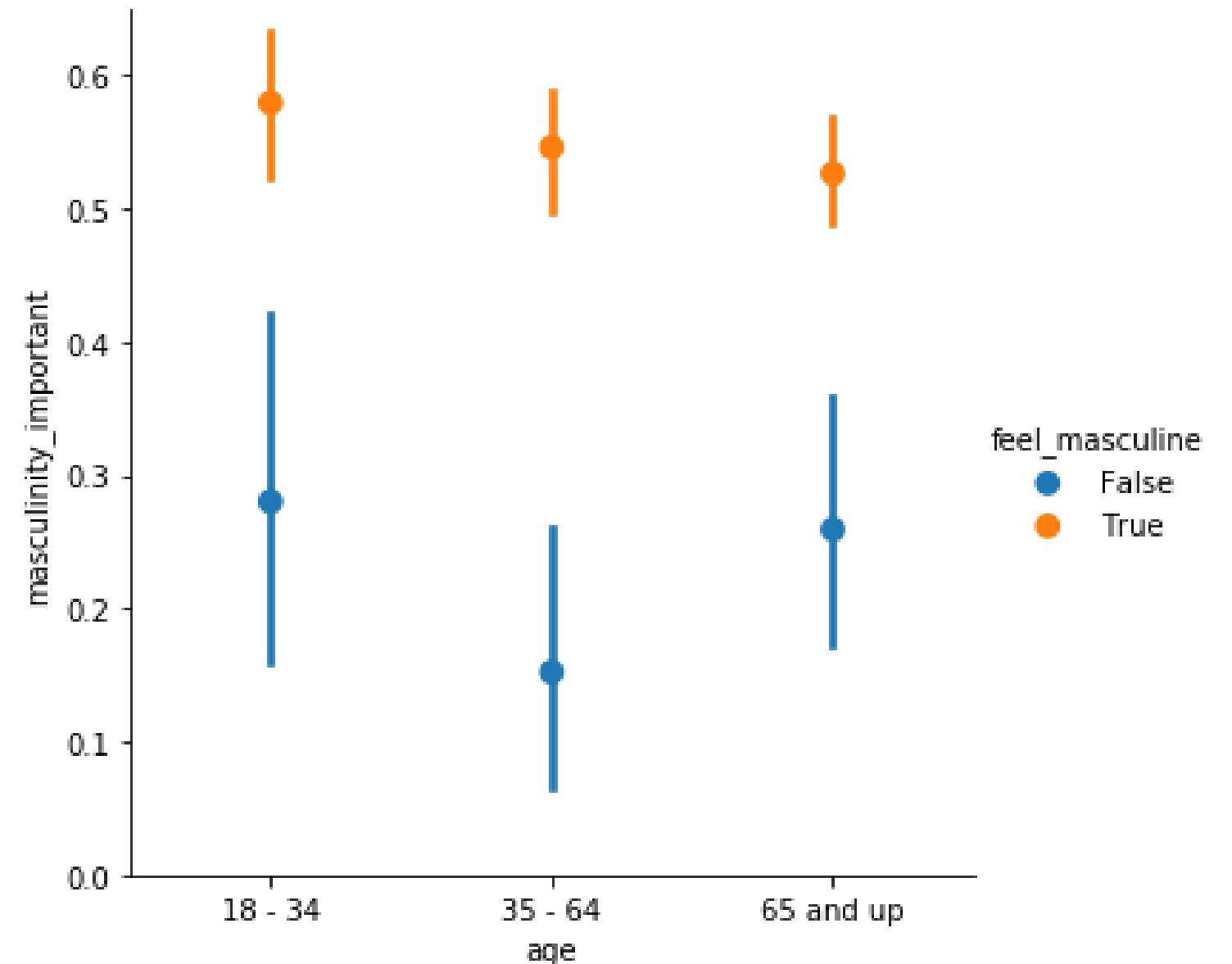


Disconnecting the points

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="age",
            y="masculinity_important",
            data=masculinity_data,
            hue="feel_masculine",
            kind="point",
            linestyle="none")

plt.show()
```

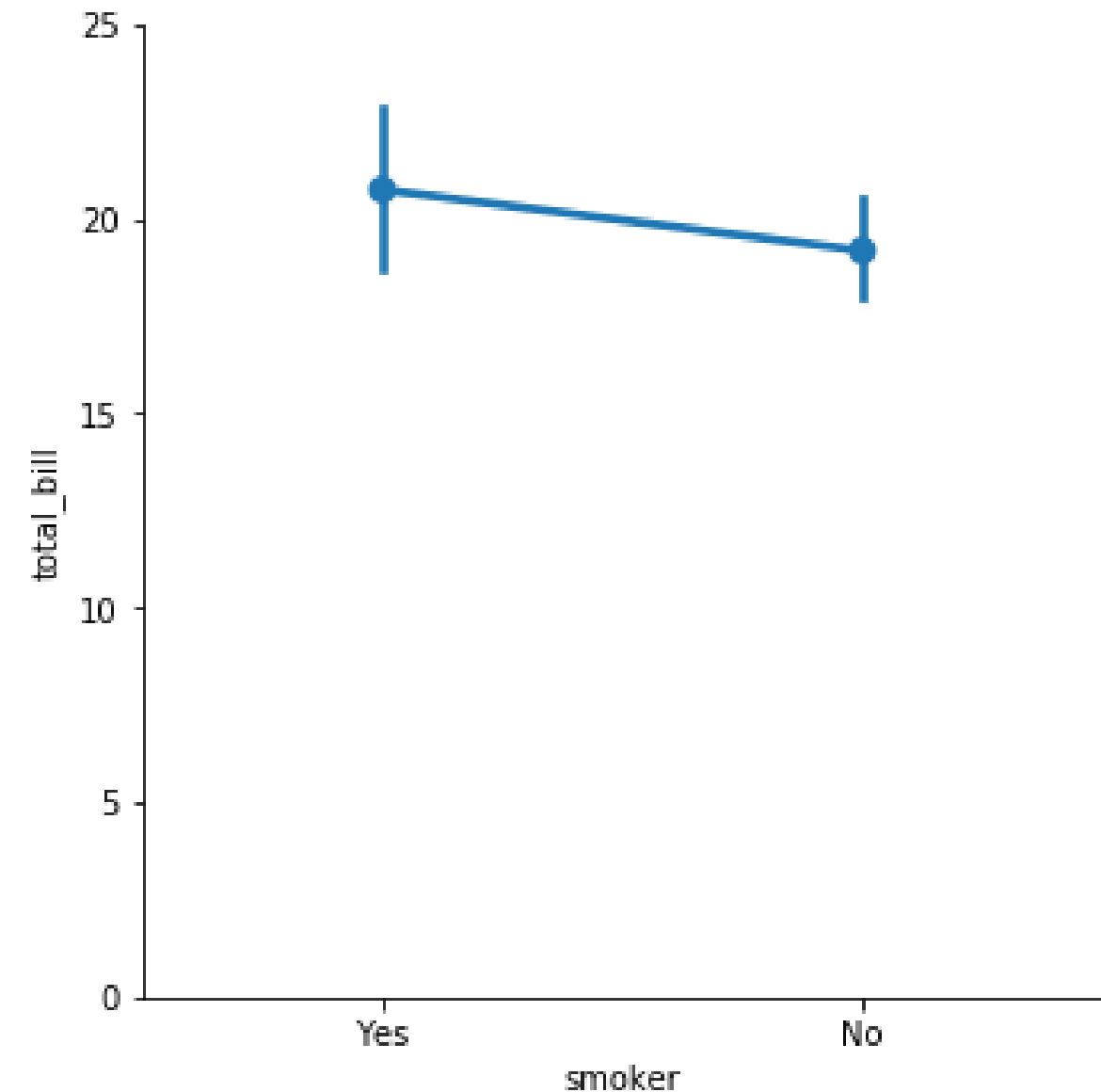


Displaying the median

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="smoker",
            y="total_bill",
            data=tips,
            kind="point")

plt.show()
```



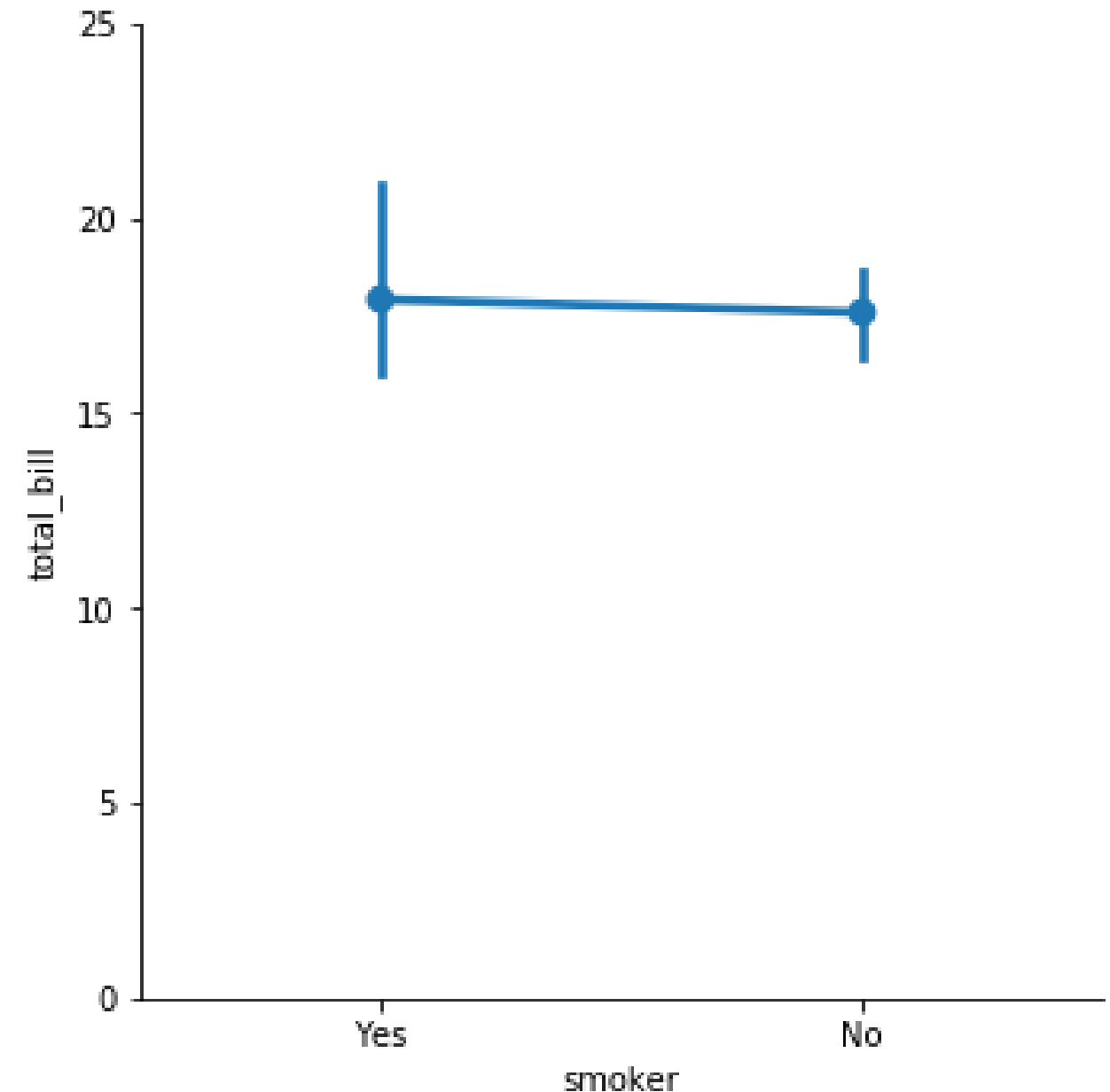
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Displaying the median

```
import matplotlib.pyplot as plt
import seaborn as sns
from numpy import median

sns.catplot(x="smoker",
            y="total_bill",
            data=tips,
            kind="point",
            estimator=median)

plt.show()
```



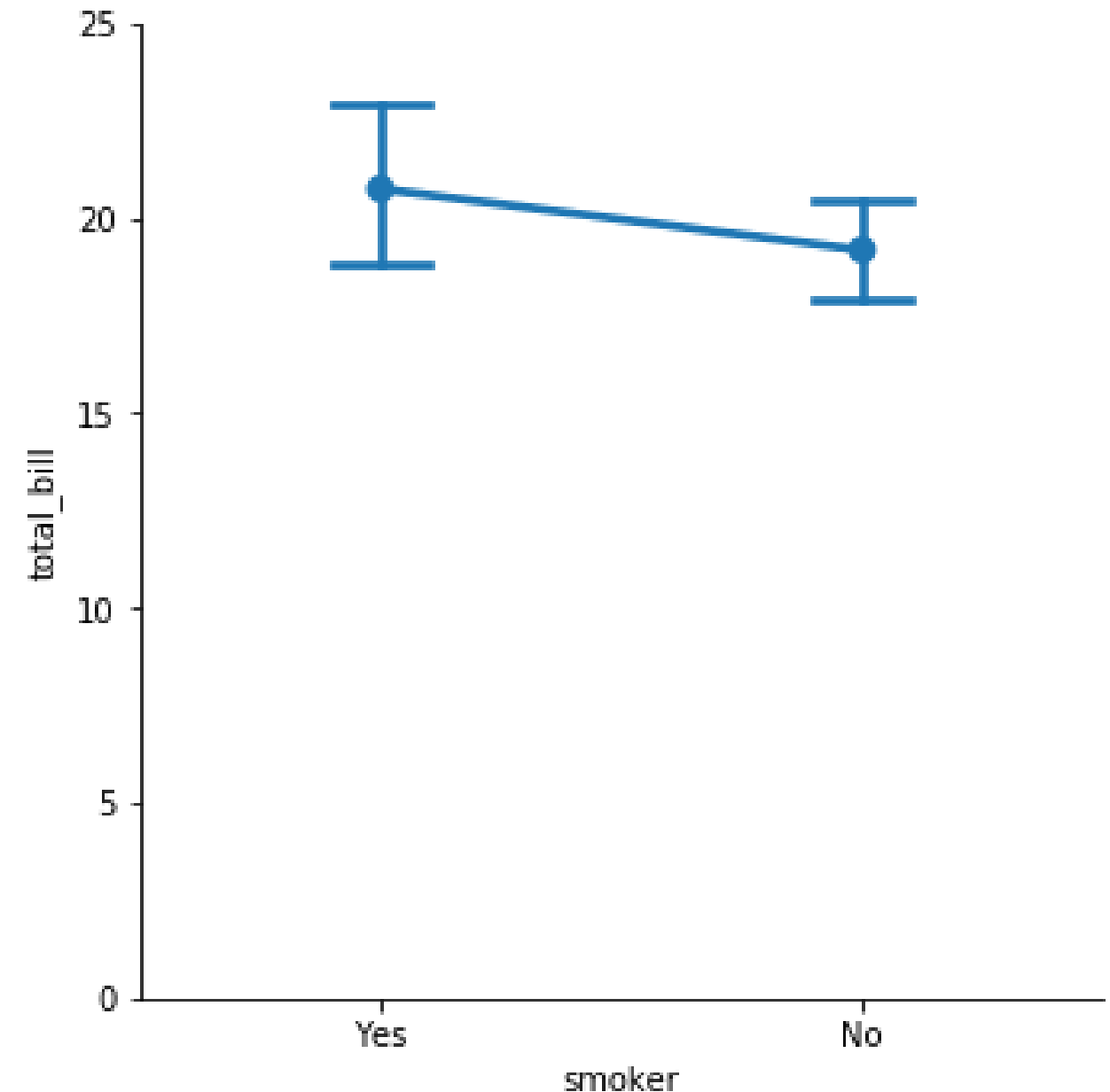
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Customizing the confidence intervals

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="smoker",
            y="total_bill",
            data=tips,
            kind="point",
            capsize=0.2)

plt.show()
```



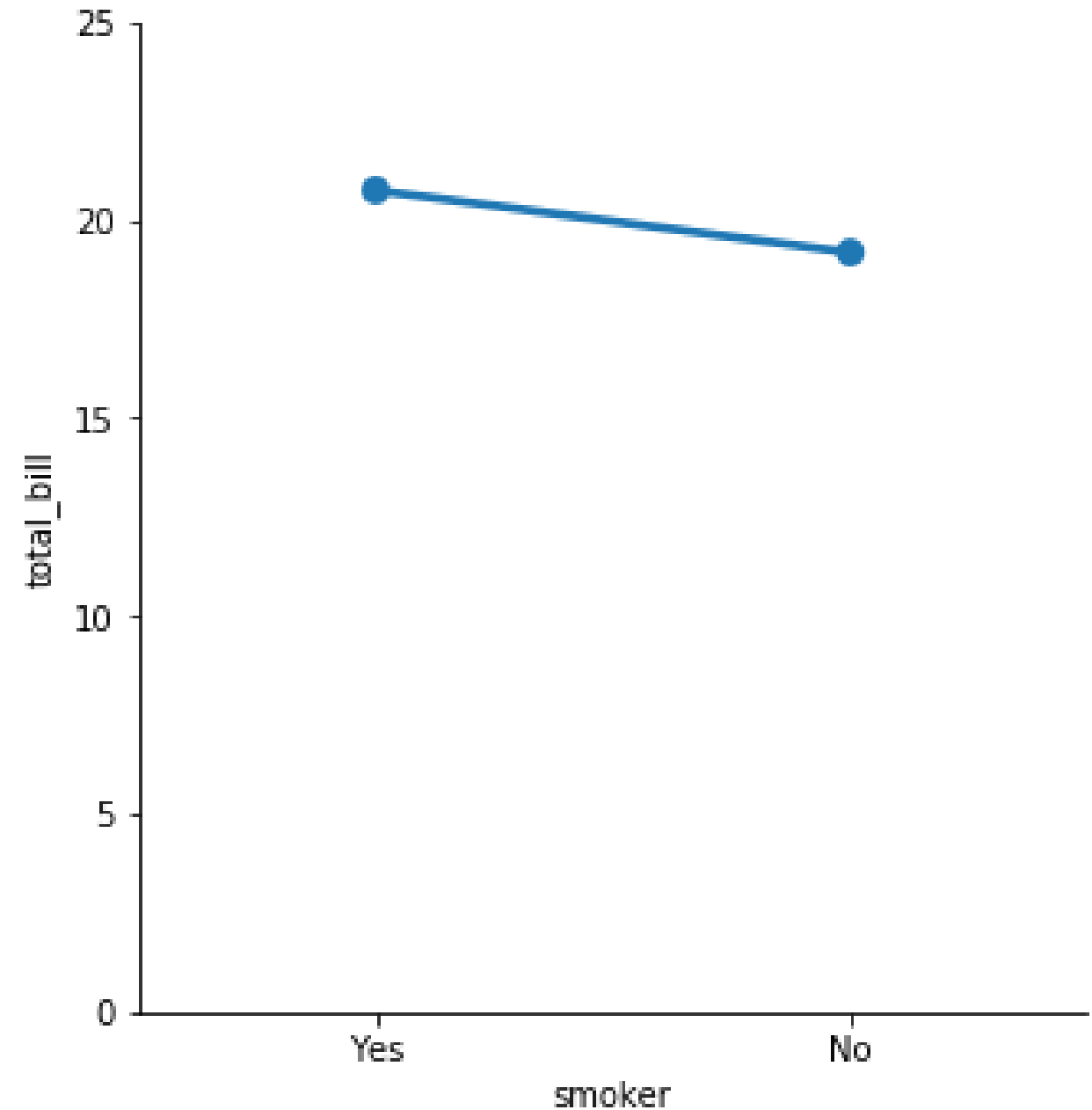
¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Turning off confidence intervals

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.catplot(x="smoker",
            y="total_bill",
            data=tips,
            kind="point",
            errorbar=None)

plt.show()
```



¹ Waskom, M. L. (2021). seaborn: statistical data visualization. <https://seaborn.pydata.org/>

Let's practice!

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